PROBLEM 03 | Intrinsic Kinematics

Kinematics, Astroid

Original statement · Problems.pdf, p. 6

A point particle describes the astroid:

$$x = a\cos^3\phi, \qquad y = a\sin^3\phi$$

with a motion whose tangential acceleration is constant, $a_t = k$. If the point is initially at rest at $\phi = 0$, compute the evolution of the position as a function of time while the particle is in one of the smooth segments of the astroid. *(Hint: $ds = \quad dx^2 + dy^2$).*

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle P moves along a hypocycloid with four cusps (an astroid), parameterized by angle ϕ.
- **Hypotheses:** The motion starts from rest at $\phi(0) = 0$ with constant tangential acceleration $a_t = k = \text{const} > 0$. The particle remains within the first quadrant segment $0 \le \phi \le \pi/2$.
- **Degrees of Freedom:** $\text{CDOF} = 1$.
- **Parameters & Units:** Astroid scale a $[\text{m}]$, tangential acceleration k $[\text{m/s}^2]$, time t $[\text{s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$.
- **Parametric Equations of the Astroid:**

$$x(\phi) = a\cos^3\phi, \quad y(\phi) = a\sin^3\phi$$

- **Metric Arc Length Differential ds:**

Differentiating with respect to parameter ϕ:

$$dx = \frac{dx}{d\phi}d\phi = 3a\cos^2\phi(-\sin\phi)d\phi = -3a\cos^2\phi\sin\phi\,d\phi$$

$$dy = \frac{dy}{d\phi}d\phi = 3a\sin^2\phi(\cos\phi)d\phi = 3a\sin^2\phi\cos\phi\,d\phi$$

The squared differential arc length is:

$$ds^2 = dx^2 + dy^2 = 9a^2\cos^4\phi\sin^2\phi\,d\phi^2 + 9a^2\sin^4\phi\cos^2\phi\,d\phi^2$$

Factoring out $9a^2\cos^2\phi\sin^2\phi$:

$$ds^2 = 9a^2\cos^2\phi\sin^2\phi(\cos^2\phi + \sin^2\phi)d\phi^2 = 9a^2\cos^2\phi\sin^2\phi\,d\phi^2$$

Taking the positive square root for $0 \le \phi \le \pi/2$:

$$ds = 3a\sin\phi\cos\phi\,d\phi = \frac{3a}{2}\sin(2\phi)\,d\phi$$

PHASE 3

Step-by-Step Mathematical Deduction

Step 1: Intrinsic Kinematics from Constant Tangential Acceleration $a_t = k$

> **Pedagogical Justification:** Tangential acceleration is the time derivative of speed: $a_t = \frac{dv}{dt} = \ddot{s}$. Integrating once with initial condition $v(0) = 0$:

$$v(t) = \dot{s}(t) = \int_0^t a_t \, dt' = \int_0^t k \, dt' = kt$$

- **Second Integration for Arc Length $s(t)$:**

Integrating $v(t) = \dot{s}(t)$ with initial condition $s(0) = 0$ (at $\phi = 0$):

$$s(t) = \int_0^t kt' \, dt' = \left[\frac{1}{2} k t'^2 \right]_0^t = \frac{1}{2} k t^2$$

Step 2: Arc Length as a Function of Parameter ϕ

> **Pedagogical Justification:** We integrate the geometric differential $ds = 3a \sin u \cos u \, du$ from the origin $\phi = 0$ to current ϕ:

- **Explicit Substitution Integration:**

Let $w = \sin u \implies dw = \cos u \, du$. When $u = 0, w = 0$; when $u = \phi, w = \sin \phi$:

$$s(\phi) = \int_0^{\phi} 3a \sin u \cos u \, du = 3a \int_0^{\sin \phi} w \, dw = 3a \left[\frac{w^2}{2} \right]_0^{\sin \phi} = \frac{3a}{2} \sin^2 \phi$$

Step 3: Equating Kinematic and Geometric Arc Length

> **Pedagogical Justification:** The distance traveled along the curve $s(t)$ must equal the geometric arc length $s(\phi)$:

$$\frac{3a}{2}\sin^2\phi = \frac{1}{2}kt^2$$

Multiplying both sides by $\frac{2}{3a}$:

$$\sin^2\phi = \frac{kt^2}{3a}$$

Taking the positive square root (since $0 \le \phi \le \pi/2$):

$$\sin\phi = \sqrt{\frac{k}{3a}}\,t$$

Applying the inverse sine function:

$$\phi(t) = \arcsin\left(\sqrt{\frac{k}{3a}}\,t\right)$$

(Compare with the official solution key (Problems.pdf, Solution keys section): $\phi = \arcsin\sqrt{\frac{k}{3a}}t^2 = \arcsin(\sqrt{\frac{k}{3a}}t)$.)

Step 4: Evolution of the Cartesian Position as a Function of Time

> **Pedagogical Justification:** Substituting $\phi(t)$ into the parametric definitions of the astroid:

Using trigonometric identities: $\sin\phi = \sqrt{\frac{k}{3a}}t$ and $\cos\phi = \sqrt{1-\sin^2\phi} = \sqrt{1-\frac{kt^2}{3a}}$:

$$x(t) = a\cos^3\phi = a\left(1 - \frac{kt^2}{3a}\right)^{3/2}$$

$$y(t) = a\sin^3\phi = a\left(\frac{kt^2}{3a}\right)^{3/2} = a\left(\frac{k}{3a}\right)^{3/2} t^3$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional Check:**
- In $\sin\phi = \quad \frac{k}{3a}t$:

$$\left[\quad \frac{k}{3a}\,t\right] = \quad \frac{\mathrm{m/s^2}}{\mathrm{m}}\,\mathrm{s} = \quad \mathrm{s^{-2}\,s = s^{-1}\cdot s} = 1 \quad \text{(Dimensionless argument for arcsin! ?)}$$

- **Physical Limits:**
- At $t = 0$: $\phi = 0 \implies x(0) = a, y(0) = 0$ (starts at the right cusp).
- The particle reaches the top cusp $\phi = \pi/2$ when $\sin\phi = 1$:

$$\frac{k}{3a}t_{\text{cusp}} = 1 \implies t_{\text{cusp}} = \quad \frac{3a}{k}$$

At t_{cusp}, $x(t_{\text{cusp}}) = 0, y(t_{\text{cusp}}) = a$.

Complete Step-by-Step Solution

Source: Problems.pdf (Page 6)